Pi & Planck

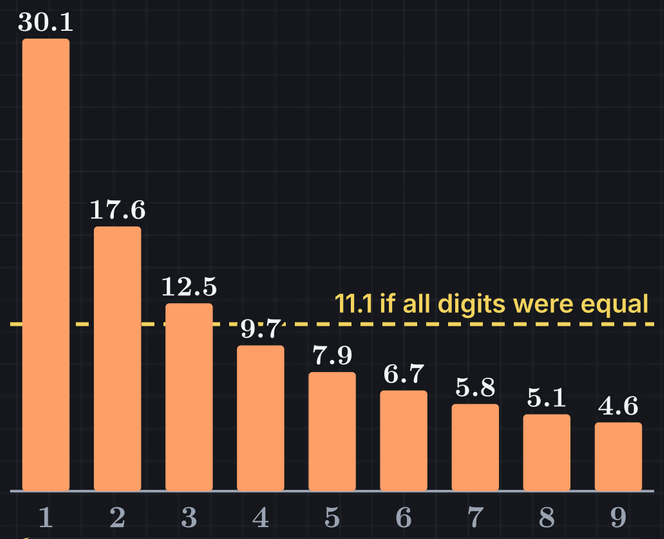
COUNTER-INTUITIVE MATH

Why So Many Numbers Start With 1

In many real datasets, about 30% of numbers start with 1 and fewer than 5% with 9. It is called Benford's law.

THE LAW

% of numbers by first digit



$$P(d) = \log_{10}(1 + 1/d)$$

d = first digit, 1 to 9

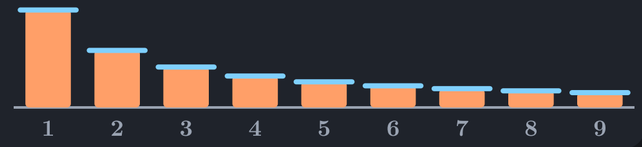
REAL SEQUENCES

counted — Benford's law

Powers of 2

$2, 4, 8 \ldots 2^{1000}$

30.1% start with 1



Fibonacci

first 1000 terms

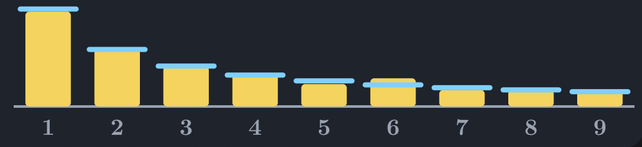
30.1% start with 1



Factorials

$1!, 2!, 3! \ldots 1000!$

29.3% start with 1



THE LOG RULER: the stretch for 1 is the longest



Spacing is by logarithm, so 1 to 2 takes 30% of the length from 1 to 10.

WHY IT HAPPENS

Growth of 7% a year (years)

1000 to 2000 — 10.2

9000 to 10 000 — 1.6

1000 to 2000 is a doubling; 9000 to 10 000 is only 11% more.

1881: worn pages

Simon Newcomb noticed that the early pages of logarithm tables wore out faster than the later ones.

1938: 20 datasets

Frank Benford tested 20,229 numbers, from river areas to atomic weights, and digit 1 led in them too.

30.1% of numbers in many real datasets begin with 1

4.6% begin with 9

6.6× more likely to begin with 1 than with 9

Sources: S. Newcomb, American Journal of Mathematics 4 (1881) 39–40 · F. Benford, Proc. American Philosophical Society 78 (1938) 551–572 · T. Hill, Statistical Science 10 (1995) 354–363 · Sequences computed for this graphic

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